

SQL Experiment-1

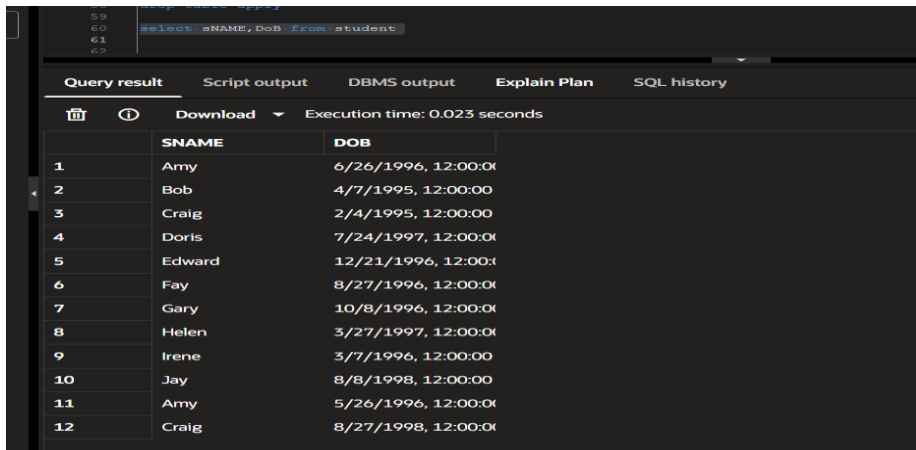
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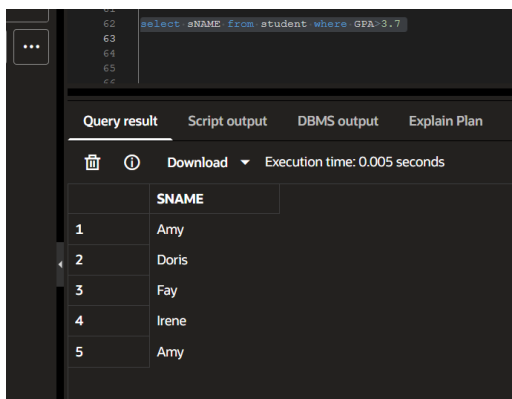
1. List the student name, dob from student table.



The screenshot shows a SQL query result in a dark-themed interface. The query is `select sname, dob from student`. The result table has two columns: **SNAME** and **DOB**. It contains 12 rows of data. The execution time is 0.023 seconds.

	SNAME	DOB
1	Amy	6/26/1996, 12:00:00
2	Bob	4/7/1995, 12:00:00
3	Craig	2/4/1995, 12:00:00
4	Doris	7/24/1997, 12:00:00
5	Edward	12/21/1996, 12:00:00
6	Fay	8/27/1996, 12:00:00
7	Gary	10/8/1996, 12:00:00
8	Helen	3/27/1997, 12:00:00
9	Irene	3/7/1996, 12:00:00
10	Jay	8/8/1998, 12:00:00
11	Amy	5/26/1996, 12:00:00
12	Craig	8/27/1998, 12:00:00

2. List the name of student scoring more than 3.7 in GPA.



The screenshot shows a SQL query result in a dark-themed interface. The query is `select sname from student where GPA > 3.7`. The result table has one column: **SNAME**. It contains 5 rows of data. The execution time is 0.005 seconds.

	SNAME
1	Amy
2	Doris
3	Fay
4	Irene
5	Amy

3. List the name of student whose High School size is atleast 1000 and born after 1996.

[Hint: check DoB greater than 31st December, 1996]

```

64 select sName from student where sizeHS=100 AND DoB>'31-Dec-1996'
65
66
67
68

```

	SNAME
1	Doris
2	Helen
3	Jay
4	Craig

4. List the name of student who are scoring GPA in between 2.9 and 3.9

```

66 select sName from student where GPA between 2.9 and 3.9
67
68
69

```

	SNAME
1	Amy
2	Bob
3	Craig
4	Doris
5	Edward
6	Fay
7	Gary
8	Helen
9	Irene
10	Jay
11	Amy
12	Craig

5. List all the details of colleges who situated in MA.

```

68 select * from college where state = 'MA'
69

```

	CNAME	STATE	ENROLLMENT
1	MIT	MA	10000
2	Harvard	MA	50040

6. List the students who are scored more than 2.0 but less than 3.5.

70

```
select * from student where GPA>2.0 and gpa<3.5
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.004 seconds

	SID	SNAME	GPA	SIZEHS	DOB
1	567	Edward	2.9	2000	12/21/1996, 12:00:00
2	789	Gary	3.4	800	10/8/1996, 12:00:00
3	765	Jay	2.9	1500	8/8/1998, 12:00:00
4	543	Craig	3.4	2000	8/27/1998, 12:00:00

7. List the students who have born after 1st Jul 96 in the order of the Date of Birth.

72

```
select * from student where DoB>'01-Jul-1996' order by DoB
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.004 seconds

	SID	SNAME	GPA	SIZEHS	DOB
1	678	Fay	3.8	200	8/27/1996, 12:00:00
2	789	Gary	3.4	800	10/8/1996, 12:00:00
3	567	Edward	2.9	2000	12/21/1996, 12:00:00
4	987	Helen	3.7	800	3/27/1997, 12:00:00
5	456	Doris	3.9	1000	7/24/1997, 12:00:00
6	765	Jay	2.9	1500	8/8/1998, 12:00:00
7	543	Craig	3.4	2000	8/27/1998, 12:00:00

8. List the sID, cName, decision of applications that are accepted.

74

```
select sID,cNAME,decision from apply where decision='Y'
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.005 seconds

	SID	CNAME	DECISION
1	123	Stanford	Y
2	123	Berkeley	Y
3	123	Cornell	Y
4	345	MIT	Y
5	345	Cornell	Y
6	678	Stanford	Y
7	987	Stanford	Y
8	987	Berkeley	Y
9	876	MIT	Y
10	765	Stanford	Y
11	765	Cornell	Y

9. List the sID, cName of applications which are filled at Stanford.

76 `select sID,cNAME from apply where cNAME = 'Stanford'`

Query result Script output DBMS output Explain Plan

Download Execution time: 0.003 seconds

	SID	CNAME
1	123	Stanford
2	123	Stanford
3	678	Stanford
4	987	Stanford
5	876	Stanford
6	765	Stanford

10. List the colleges that that has enrollment greater than 10001.

78 `select * from college where enrollment > 10001`

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.005 seconds

	CNAME	STATE	ENROLLMENT
1	Stanford	CA	15000
2	Berkeley	CA	36000
3	Cornell	NY	21000
4	Harvard	MA	50040

11. List the colleges not in California.

80 `select * from college where state != 'CA'`

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.003 seconds

	CNAME	STATE	ENROLLMENT
1	MIT	MA	10000
2	Cornell	NY	21000
3	Harvard	MA	50040

12. List names of all student who came from high school having size greater than 17000 and scored GPA less than 3.8.

82 `select sName from student where sizeHS > 17000 and gpa < 3.8`

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.003 seconds

SNAME
No items to display.

13. Display the description of the Student table.

```
84 describe student
```

Query result Script output

SQL> describe student

Name	Null?	Type
SID		NUMBER(38)
SNAME		VARCHAR2(10)
GPA		NUMBER(2,1)
SIZEHS		NUMBER(38)
DOB		DATE

14. Display the details of all students.

```
86 select * from student
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.002 seconds

	SID	SNAME	GPA	SIZEHS	DOB
1	123	Amy	3.9	1000	6/26/1996, 12:00:00
2	234	Bob	3.6	1500	4/7/1995, 12:00:00
3	345	Craig	3.5	500	2/4/1995, 12:00:00
4	456	Doris	3.9	1000	7/24/1997, 12:00:00
5	567	Edward	2.9	2000	12/21/1996, 12:00:00
6	678	Fay	3.8	200	8/27/1996, 12:00:00
7	789	Gary	3.4	800	10/8/1996, 12:00:00
8	987	Helen	3.7	800	3/27/1997, 12:00:00
9	876	Irene	3.9	400	3/7/1996, 12:00:00
10	765	Jay	2.9	1500	8/8/1998, 12:00:00
11	654	Amy	3.9	1000	5/26/1996, 12:00:00
12	543	Craig	3.4	2000	8/27/1998, 12:00:00

15. Display unique majors.

88	<code>select distinct major from apply</code>
Query result Script output DBMS output Download Execution time: 0.004 seconds	
	MAJOR
1	CS
2	EE
3	biology
4	bioengineering
5	history
6	marine biology
7	psychology

16. List the student names those are having three characters in their Names.

90	<code>select sNAME from student where sNAME like '___'</code>
Query result Script output DBMS output Explain Plan Download Execution time: 0.004 seconds	
	SNAME
1	Amy
2	Bob
3	Fay
4	Jay
5	Amy

17. List the student names those are starting with 'H' and with five characters.

92	<code>select sName from student where sName like 'H_____'</code>
Query result Script output DBMS output Explain Plan Download Execution time: 0.003 seconds	
	SNAME
1	Helen

18. List the student names those are having third character and fifth char. must be 'e'.

94	<code>select sName from student where sName like '___e_'</code>
Query result Script output DBMS output Explain Plan Download Execution time: 0.003 seconds	
	SNAME
1	Irene

19. List the student names ending with 'y'.

96 `select sName from student where sName like 'Ay'`

Query result Script output DBMS output Explain Plan

Download Execution time: 0.003 seconds

	SNAME
1	Amy
2	Fay
3	Gary
4	Jay
5	Amy

20. List the Students in the order of their GPA.

98 `select * from student order by GPA`

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.004 seconds

	SID	SNAME	GPA	SIZEHS	DOB
1	765	Jay	2.9	1500	8/8/1998, 12:00:00
2	567	Edward	2.9	2000	12/21/1996, 12:00:00
3	543	Craig	3.4	2000	8/27/1998, 12:00:00
4	789	Gary	3.4	800	10/8/1996, 12:00:00
5	345	Craig	3.5	500	2/4/1995, 12:00:00
6	234	Bob	3.6	1500	4/7/1995, 12:00:00
7	987	Helen	3.7	800	3/27/1997, 12:00:00
8	678	Fay	3.8	200	8/27/1996, 12:00:00
9	876	Irene	3.9	400	3/7/1996, 12:00:00
10	123	Amy	3.9	1000	6/26/1996, 12:00:00
11	654	Amy	3.9	1000	5/26/1996, 12:00:00
12	456	Doris	3.9	1000	7/24/1997, 12:00:00

21. List the details of the students in order of the ascending of GPA and descending of DoB.

100 `select * from student order by GPA asc,DOB desc`

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.005 seconds

	SID	SNAME	GPA	SIZEHS	DOB
1	765	Jay	2.9	1500	8/8/1998, 12:00:00
2	567	Edward	2.9	2000	12/21/1996, 12:00:00
3	543	Craig	3.4	2000	8/27/1998, 12:00:00
4	789	Gary	3.4	800	10/8/1996, 12:00:00
5	345	Craig	3.5	500	2/4/1995, 12:00:00
6	234	Bob	3.6	1500	4/7/1995, 12:00:00
7	987	Helen	3.7	800	3/27/1997, 12:00:00
8	678	Fay	3.8	200	8/27/1996, 12:00:00
9	456	Doris	3.9	1000	7/24/1997, 12:00:00
10	123	Amy	3.9	1000	6/26/1996, 12:00:00
11	654	Amy	3.9	1000	5/26/1996, 12:00:00
12	876	Irene	3.9	400	3/7/1996, 12:00:00

22. List the sIDs of student who apply in either 'Stanford', 'Cornell' or 'MIT' college.

102 `select sID from apply where cNAME IN ('Stanford','Cornell','MIT')`

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.003 seconds

	SID
1	123
2	123
3	123
4	345
5	345
6	345
7	345
8	678
9	987
10	876
11	876
12	876
13	765
14	765
15	765
16	543

23. Delete all applications filled at Stanford (Choose table wisely)


```
104 delete from apply where cNAME='Stanford'
```

Query result Script output DBMS output Exp

SQL> delete from apply where cNAME='Stanford'

6 rows deleted.

Elapsed: 00:00:00.003

24. Delete the college Stanford from college table.

```
106 delete from college where cNAME = 'Stanford'
```

Query result Script output DBMS output Explain Plan

SQL> delete from college where cNAME = 'Stanford'

1 row deleted.

Elapsed: 00:00:00.003

25. Modify the GPA of all students by giving 10% raise in their GPA.

```
SQL> update student set GPA = GPA*1.10
```

12 rows updated.

Elapsed: 00:00:00.003

26. Increment the GPA of the students by 1.5 whose GPA is less than 3.5 and belong to High School having size greater than 1500.

```
SQL> update student set GPA = GPA+1.5 where GPA<3.5 and sizeHS>1500
```

1 row updated.

Elapsed: 00:00:00.003

27. Delete the students who have scored less than 3.2 GPA.

```
SQL> delete from student where GPA<3.2
```

0 rows deleted.

Elapsed: 00:00:00.003

